

```
clear all
```

Primero se define en New Script la función

```
function dydt = vdp1(t,y)
```

```
%VDP1 Evaluate the van der Pol ODEs for mu = 1
```

```
%
```

```
% See also ODE113, ODE23, ODE45.
```

```
% Jacek Kierzenka and Lawrence F. Shampine
```

```
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```

```
dydt = [y(2); (1-y(1)^2)*y(2)-y(1)];
```

```
[t,y]=ode45(@vdp1,[0,20],[2; 0])
```

```
t =
```

```
0
0.000025118864315
0.000050237728630
0.000075356592945
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0.000351664100411
0.000477258421987
0.000602852743562
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0.006254597214459
0.009394455253846
0.012534313293233
0.015674171332620
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0.047072751726490
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0.129953180498351
0.181435028876342
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0.284398725632325
0.348473667842497
0.412548610052669
0.476623552262840
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0.709741546383414
0.794263072338615
0.878784598293816
```

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y =

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```

plot(t,y(:,1),'-o',t,y(:,2),'-o')
title('Solution of van der Pol Equation (\mu = 1) with ODE45');
xlabel('Time t');
ylabel('Solution y');
legend('y_1','y_2')

```


